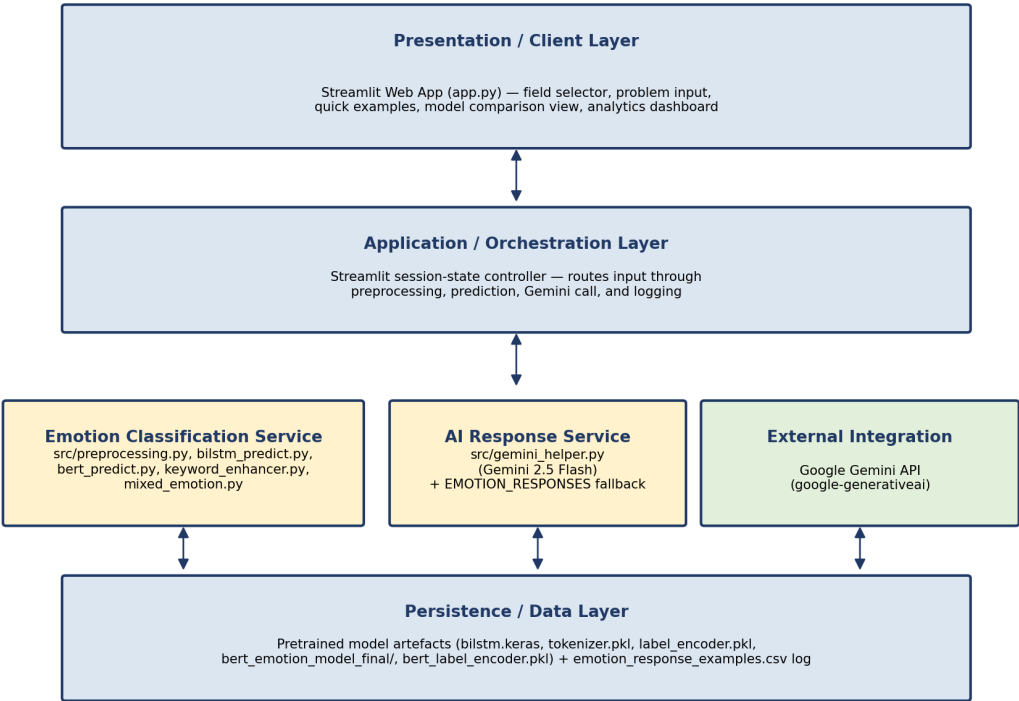


Solution Architecture

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	5 Marks

Solution Architecture Diagram

The application follows a layered architecture: a Streamlit presentation layer, an application/orchestration layer that coordinates prediction and AI response generation, and a persistence layer holding the trained model artefacts and the interaction log.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the study-problem input form, quick examples, model comparison panels, AI response card, and the analytics dashboard (tabs for Emotions / Fields / Summary).	Streamlit, Plotly Express

Component Name	Description / Role in Architecture	Technologies Used
Emotion Classification Service	Cleans the raw text, runs it through the BiLSTM and BERT models in parallel, applies keyword boosting and mixed-emotion thresholding, and returns a unified prediction schema.	TensorFlow/Keras, PyTorch, HuggingFace Transformers, NLTK
AI Response Service	Builds a structured prompt from field/problem/emotion/confidence, calls the Gemini API, and returns the generated text; falls back to a curated EMOTION_RESPONSES dictionary on failure.	google-generativeai (Gemini 2.5 Flash), python-dotenv
Persistence / Data Layer	Stores pretrained model weights, tokenizers and label encoders used at inference time, and appends every analysed interaction to a CSV log.	Pickle, Keras .keras format, HuggingFace model directory, pandas CSV